

Solving Linear Systems

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)
Based on the lecture note uploaded on SPeCTRUM

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Note: The techniques you learn in this section would be constantly used in some higher level courses, especially if you're planning to take SIM2002 Linear Algebra, so make sure you learn it carefully and get lots of practice at it.

Lec
22/12 (Mon) in-person. 22/12 in-person
23/12 in-person.

24/12 (Wed) ← Hybrid
25/12

29/12 (Mon) Hybrid 29/12 Hybrid
Hybrid. 30/12 Hybrid

31/12 (Wed) ← 1/1

Tut Lec.

W10 in-person

22/12 in-person

W11 hybrid.

24/12 and W11 hybrid.

* Subject to university latest instruction.

Learning Outcomes

At the end of this lecture, you should be able to:

- Know which elementary row operations are available and perform them.
- Understand what is the leading row, leading entry, leading column, row-echelon form, and reduced row-echelon form.
- Performs the back-substitution to get the solutions of the system of linear equations.
- Understand what is the pivot element, pivot row, and pivot column.
- Understand and perform the Gaussian Elimination and Gaussian-Jordan Elimination to obtain a matrix in row-echelon form and reduced row-echelon form.
- Able to tell the number of solutions from a matrix in row-echelon form and understand why.
- Know what is a homogeneous system of linear equations and able to find its solution set.

Elementary Row Operations

A matrix A is said to be **row equivalent** to a matrix B if B can be obtained from A by a finite sequence of the following operations called **elementary row operations**:

[E1]: Interchanging the i^{th} row and the j^{th} row: $R_i \leftrightarrow R_j$

[E2]: Multiplication of a row by a nonzero constant λ : $R_i \rightarrow \lambda R_i$,
 $\lambda \in \mathbb{R} \setminus \{0\}$

[E3]: Addition of a constant multiple of one row to another:
 $R_i \rightarrow R_i + \beta R_j$, $\beta \in \mathbb{R}$

Elementary Row Operations

Example:

$$[E1]: \begin{bmatrix} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ -1 & 3 & 4 & 4 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} -1 & 3 & 4 & 4 \\ 3 & -1 & 1 & 6 \\ -1 & 1 & 2 & 2 \end{bmatrix}$$

$$[E2]: \begin{bmatrix} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ -1 & 3 & 4 & 4 \end{bmatrix} \xrightarrow{R_3 \rightarrow -3R_3} \begin{bmatrix} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ 3 & -9 & -12 & -12 \end{bmatrix}$$

$$[E3]: \begin{bmatrix} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ -1 & 3 & 4 & 4 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + 3R_3} \begin{bmatrix} -1 & 1 & 2 & 2 \\ 0 & 8 & 13 & 18 \\ -1 & 3 & 4 & 4 \end{bmatrix}$$

Elementary Row Operations

If augmented matrices of two systems of linear equations are row equivalent, then the two systems have the same set of solutions.

Example:

Consider this system of linear equations:

$$\begin{array}{rcl} -x_1 + x_2 + 2x_3 & = & 2, \\ 3x_1 - x_2 + x_3 & = & 6, \\ -x_1 + 3x_2 + 4x_3 & = & 4. \end{array}$$

$$\begin{array}{l} 3 \times \textcircled{1} - \textcircled{3} \\ \textcircled{2} + \textcircled{4} \end{array}$$

[E1]: $R_1 \leftrightarrow R_3$

$$\begin{array}{l} -x_1 + 3x_2 + 4x_3 = 4 \\ 3x_1 - x_2 + x_3 = 6 \\ -x_1 + x_2 + 2x_3 = 2. \end{array}$$

[E2]: $R_3 \rightarrow -R_3$

$$\begin{array}{l} -x_1 + 3x_2 + 4x_3 = 4 \\ 3x_1 - x_2 + x_3 = 6 \\ x_1 - 3x_2 - 4x_3 = -4 \end{array}$$

[E3]: $R_2 \rightarrow R_2 + 3R_3$

$$\begin{array}{l} -11x_1 + 11x_2 + 2x_3 = 2 \\ 8x_2 + 13x_3 = 18 \\ -x_1 + 3x_2 + 4x_3 = 4 \end{array}$$

Leading Rows and Entries

A **leading row** is a nonzero row (one or more entries are not zero).

A **leading entry** is the leftmost nonzero entry in a leading row.

A **leading column** is a column containing a leading entry.

Example:

Consider the following matrix.

$$\begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Row 1 is a leading row with leading entry 1

Row 2 is a non-leading row.

Column 2 is the only leading column.

$$\begin{bmatrix} 0 & \textcircled{1} & \textcircled{2} \\ 0 & 0 & \textcircled{6} \\ 0 & 0 & \textcircled{1} \end{bmatrix}$$

Row 1 is a leading row with leading entry 1

Row 2 is a leading row with leading entry 6

Row 3 is a leading row with leading entry 1

Column 2 and Column 3 are leading column.

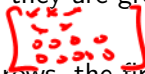
↓
contains
leading entry 1

↓
contains
leading entry 6.

Row-Echelon Form

A matrix is said to be **row-echelon form** if it has the following properties:

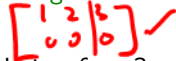
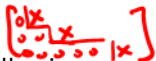
1. If there are any zero rows, then they are grouped together at the bottom of the matrix.
2. In any two successive non-zero rows, the first non-zero number (**leading entry**) in the lower row occurs farther to the right than the first non-zero number in the upper row.



All non-leading rows are at the bottom

Note: The column that contains leading entry is called a **leading column**.

Example:



Which of the following augmented matrices are in row-echelon form?

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$



$$\left[\begin{array}{cc|c} 3 & 1 & 2 \\ 0 & 0 & 1 \end{array} \right]$$



$$\left[\begin{array}{cc|c} 0 & 1 & 2 \\ 3 & -1 & 1 \end{array} \right]$$



$$\left[\begin{array}{cc|c} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

non-leading row at above



$$\left[\begin{array}{cc|c} 0 & 0 & 2 \\ 1 & 2 & 3 \end{array} \right]$$



at the left of the leading entry in the row above.

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]$$



Reduced Row-Echelon Form

A matrix is said to be **reduced row-echelon form** if it is in row-echelon form and has the following additional properties:

1. The leading entry of every nonzero row is 1.
2. Each column that contains a leading entry (**leading column**) has zeros everywhere else.

Example:

Which of the following augmented matrices are in **reduced** row-echelon form?

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

REF ✓

RREF ✓

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & 1 & 1 \end{array} \right]$$

REF ✓

RREF ✗

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 3 & 1 \end{array} \right]$$

REF ✓

RREF ✗

$$\left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right]$$

REF ✓

RREF ✓

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 0 & 0 \end{array} \right]$$

REF ✓

RREF ✓

leading entry not equal to 1.

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

REF ✓

RREF ✓

Back-Substitution

When the augmented matrix of a system of linear equations is in row-echelon form or reduced row-echelon form, we can get the solutions of the system easily using a method called **back-substitution**.

The array $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & -1 & 1 & 3 \\ 0 & 0 & 2 & 4 \end{array} \right]$ is the augmented matrix of a system of linear equations in 3 variables x_1, x_2, x_3 :

$$x_1 + 0x_2 + 0x_3 = 2 \Rightarrow x_1 = 2.$$

$$0x_1 - x_2 + x_3 = 3 \Rightarrow -x_2 + 2 = 3 \Rightarrow x_2 = -1$$

$$0x_1 + 0x_2 + 2x_3 = 4 \Rightarrow x_3 = 2$$

The system has only one solution, i.e. $x_1 = 2, x_2 = -1, x_3 = 2$.

Back-Substitution

The array $\left[\begin{array}{ccccc|c} 0 & 2 & 2 & 1 & -2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 2 & 4 \end{array} \right]$ is the augmented matrix of a system of linear equations in 5 variables x_1, x_2, x_3, x_4, x_5 :

$$2x_2 + 2x_3 + x_4 - 2x_5 = 2$$

$$x_3 + x_4 + x_5 = 3$$

$$2x_5 = 4$$

Note that x_1 does not appear in the equations, so x_1 is arbitrary (we can take x_1 be any number), $x_1 = s \in \mathbb{R}$.

From the third equation, $x_5 = 2$.

Substitute $x_5 = 2$ into second equation, $x_3 + x_4 = 1 \Rightarrow x_3 = 1 - x_4$
 $\Rightarrow x_4 = t \in \mathbb{R}, x_3 = 1 - t$

Substitute $x_3 = 1-t$, $x_4 = t$, $x_5 = 2$ into the first equation,

$$2x_2 + 2(1-t) + t - 2(2) = 2$$

$$x_2 = 2 + \frac{1}{2}t.$$

So the solution set of the system of linear equations is

$$\left\{ \begin{pmatrix} s \\ 2 + \frac{1}{2}t \\ 1-t \\ t \\ 2 \end{pmatrix}, s, t \in \mathbb{R} \right\},$$

and this system has infinitely many solutions.

Back-Substitution

The array $\left[\begin{array}{cccc|c} 1 & -1 & 0 & 3 & -2 \\ 0 & 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$ is the augmented matrix of a system of linear equations in 4 variables x_1, x_2, x_3, x_4 :

$$x_1 - x_2 + 3x_4 = -2$$

$$x_3 + 2x_4 = 5$$

$$0 = 0$$

Column 2 and Column 4 are non-leading columns.

Taking $x_2 = s$ and $x_4 = t$, where s, t are arbitrary parameters

The solution set is

$$\left\{ \begin{pmatrix} -2 + s - 3t \\ s \\ 5 - 2t \\ t \end{pmatrix}, s, t \in \mathbb{R} \right\}.$$

Back-Substitution

The array $\left[\begin{array}{cc|c} 3 & 1 & 4 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{array} \right]$ is the augmented matrix of a system of linear equations in 2 variables x_1, x_2 :

$$3x_1 + x_2 = 4$$

$$2x_2 = 1$$

$$0 = 1 \text{ (A contradiction)}$$

This system is inconsistent and has no solution.

Pivots

A **pivot element** is a nonzero entry in the first (leftmost) nonzero column ^(pivot column).

A **pivot row** is the row containing a pivot element.

A **pivot column** is a ~~row~~ ^{column} containing a pivot element.

Example:

Consider the following matrix.

$$\begin{bmatrix} 0 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 2 & 4 \end{bmatrix}$$

^{↖ ↙}
↑
pivot column

$$\begin{bmatrix} 0 & 0 & 2 \\ 0 & 2 & 4 \end{bmatrix}$$

Column 2 is a pivot column.

1 and 2 (in column 2) are the pivot elements

Row 2 and Row 3 are pivot rows.

Gaussian Elimination

To reduce a matrix to a row-echelon form by some elementary row operations.

Step 1: Locate a pivot element. If one of these leftmost nonzero entries is 1, we pick this as our pivot, otherwise, we usually pick the first non-zero entry.

Step 2: Interchange the top row with the pivot row we pick in Step 1, if necessary, to bring the pivot element we pick in Step 1 to the top row.

Step 3: Add a multiple of the pivot row to the rows below it so that all other entries in the pivot column becomes zeros. *$R_j \rightarrow R_j + \lambda R_1$ so that pivot element in R_j becomes 0.*

Step 4: Cover the top row in the matrix and repeat Step 1 to Step 3 to the submatrix that remains. Continue in this way until the entire matrix is in row-echelon form.

Gaussian Elimination

Example:

Use Gaussian Elimination to reduce the matrix

→ $\begin{bmatrix} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{bmatrix}$ to a row-echelon form.

Step 1: Choose 1 in the entry (2,1) be our pivot element.

Step 2: Interchange R_2 with R_1 so that '1' in Step 1 is in the top row.

$$\begin{bmatrix} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{bmatrix}$$

Gaussian Elimination

Step 3: Use an elementary row operation to the third row so that the entry $(3,1)$ is zero.

$$\begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + 2R_1} \begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 3 & 6 & 9 & -12 \end{bmatrix}$$

Step 4: Cover the top row and repeat steps 1-3.

$$\begin{bmatrix} \text{1} & \text{2} & \text{4} & \text{5} & \text{3} & \text{-9} \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 3 & 6 & 9 & -12 \end{bmatrix}$$

↑
pivot column.
(blue colour)

↑
not yet
in REF

Step 4.1 : The 2 in the entry (2,3) is the chosen pivot element

Step 4.2 : No action is needed.

Step 4.3 : Use an elementary row operation on the third row so that entry (3,3) is zero.

$$\begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 2 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & \textcircled{3} & 6 & 9 & -12 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - \frac{3}{2}R_2} \begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{bmatrix}$$

↑
The matrix is now in row-echelon form.

Gaussian-Jordan Elimination

(Gauss-Jordan Elimination)

To reduce a matrix to a reduced row-echelon form by some elementary row operations.

Using Steps 1-4 of Gaussian Elimination until the matrix is in row-echelon form.

Step 5: Multiply a suitable constant to each row so that the leading entries become 1.

Step 6: Beginning with the last leading row and working upward, apply some elementary row operations $R_i \rightarrow R_i + \beta R_j$, $\beta \in \mathbb{R}$ to the row above to introduce zero above the leading entries.

Gaussian-Jordan Elimination

Example:

Use Gaussian-Jordan Elimination to reduce the matrix

$$\begin{bmatrix} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{bmatrix} \text{ to a reduced row-echelon form.}$$

Step 1-4: $\begin{bmatrix} \textcircled{1} & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & \textcircled{2} & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & \textcircled{6} & -24 \end{bmatrix}$ (last example)

Step 5: Multiply a suitable constant to each row so that the leading entries becomes 1.

$$\begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{bmatrix} \xrightarrow[R_3 \rightarrow \frac{1}{6}R_3]{R_2 \rightarrow \frac{1}{2}R_2} \begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix}$$

↑ ↑ ↑

Gaussian-Jordan Elimination

Step 6: Apply some elementary row operations to R_1 and R_2 so that '1' and '3' (above the leading entry '1') becomes 0.

$$\begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix} \xrightarrow[\substack{R_2 \rightarrow R_2 - R_3 \\ R_1 \rightarrow R_1 - 3R_3}]{\substack{R_2 \rightarrow R_2 - R_3 \\ R_1 \rightarrow R_1 - 3R_3}} \begin{bmatrix} 1 & 2 & 4 & 5 & 0 & 3 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix}$$

↑ ↑ ↑

Repeat step 6: Apply elementary row operations to R_1 (by using R_2) so that '4' above the leading entry '1' becomes 0.

$$\begin{bmatrix} 1 & 2 & 4 & 5 & 0 & 3 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - 4R_2} \begin{bmatrix} 1 & 2 & 0 & -3 & 0 & -29 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix}$$

RRREF.

Example

Solve the system of linear equations

$$2x_3 + 4x_4 + 2x_5 = 8,$$

$$x_1 + 2x_2 + 4x_3 + 5x_4 + 3x_5 = -9,$$

$$-2x_1 - 4x_2 - 5x_3 - 4x_4 + 3x_5 = 6.$$

The augmented matrix of this system is

$$\left[\begin{array}{ccccc|c} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{array} \right]$$

↑
coefficient matrix

↑ constant.

Example

M1 : Gaussian Elimination.

Refer to previous example.

$$\text{REF: } \left[\begin{array}{ccccc|c} \textcircled{1} & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & \textcircled{2} & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & \textcircled{6} & -24 \end{array} \right]$$

Column 2 and
Column 4 are
non-leading columns
So x_2 and x_4
are arbitrary
parameters

Use back-substitution.

From the third row, $6x_5 = -24 \Rightarrow \underline{x_5 = -4}$

From the second row, $2x_3 + 4x_4 + 2x_5 = 8$

Substitute $x_5 = -4$, $\underline{x_3 = 8 - 2x_4}$

From the first row, $x_1 + 2x_2 + 4x_3 + 5x_4 + 3x_5 = -9$.

Substitute $x_3 = 8 - 2x_4$, $x_4 = x_4$, $x_5 = -4$,

Example

$$x_1 + 2x_2 + 4(8-2x_4) + 5x_4 + 3(-4) = -9$$

$$x_1 = -29 - 2x_2 + 3x_4.$$

Let $x_2 = s$, $x_4 = t$, s, t are some arbitrary parameter.

The solution set is

$$\left\{ \begin{pmatrix} -29 - 2s + 3t \\ s \\ 8 - 2t \\ t \\ -4 \end{pmatrix}, s, t \in \mathbb{R} \right\}$$

@ is the general solution

$$\left\{ \begin{array}{l} x_1 = -29 - 2s + 3t \\ x_2 = s \\ x_3 = 8 - 2t \\ x_4 = t \\ x_5 = -4 \end{array}, s, t \in \mathbb{R} \right\}$$

M2: Gauss-Jordan Elimination.

$$\text{RREF: } \left[\begin{array}{ccccc|c} 1 & 2 & 0 & -3 & 0 & -29 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{array} \right]$$

This corresponds to

$$x_1 + 2x_2 - 3x_4 = -29 \Rightarrow x_1 = -29 - 2x_2 + 3x_4$$

$$x_3 + 2x_4 = 8 \Rightarrow x_3 = 8 - 2x_4$$

$$x_5 = -4$$

Let $x_2 = s$, $x_4 = t$, s, t are arbitrary parameters.

The solution set is

$$\left\{ \begin{pmatrix} -29 - 2s + 3t \\ s \\ 8 - 2t \\ t \\ -4 \end{pmatrix}, s, t \in \mathbb{R} \right\}.$$

Example

Solve the system of linear equations

$$3x_1 + 2x_2 + 4x_3 = 1,$$

$$5x_1 - x_2 + 3x_3 = 2,$$

$$8x_1 + x_2 + 7x_3 = 4.$$

The augmented matrix of this system is

$$\left[\begin{array}{ccc|c} 3 & 2 & 4 & 1 \\ 5 & -1 & 3 & 2 \\ 8 & 1 & 7 & 4 \end{array} \right]$$

Apply Gaussian Elimination to reduce the augmented matrix into row-echelon form.

$$\left[\begin{array}{ccc|c} 3 & 2 & 4 & 1 \\ 5 & -1 & 3 & 2 \\ 8 & 1 & 7 & 4 \end{array} \right] \xrightarrow[\begin{array}{l} R_2 \rightarrow R_2 - \frac{5}{3}R_1 \\ R_3 \rightarrow R_3 - \frac{8}{3}R_1 \end{array}]{\begin{array}{ccc|c} 3 & 2 & 4 & 1 \\ 0 & \frac{13}{3} & -\frac{11}{3} & \frac{1}{3} \\ 0 & -\frac{13}{3} & -\frac{11}{3} & \frac{4}{3} \end{array}}$$

Example

$$\underline{R_3 \rightarrow R_3 - R_2} \rightarrow \left[\begin{array}{ccc|c} 3 & 2 & 4 & 1 \\ 0 & -\frac{13}{3} & -\frac{11}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 \end{array} \right]$$

This corresponding to the system

$$3x_1 + 2x_2 + 4x_3 = 1$$

$$-\frac{13}{3}x_2 - \frac{11}{3}x_3 = \frac{1}{3}$$

$$0x_1 + 0x_2 + 0x_3 = 1 \leftarrow \text{No values of } x_1, x_2, x_3 \text{ will satisfy this.}$$

The system has no solution.

(The system is inconsistent)

Example

For the following system of linear equations, find x_1, x_2, x_3 in terms of b_1, b_2, b_3

$$x_1 - 2x_2 + 3x_3 = b_1,$$

$$x_2 - 3x_3 = b_2,$$

$$-2x_1 + 3x_2 - 2x_3 = b_3.$$

The augmented matrix of this system is.

$$\begin{array}{ccc|c} x_1 & x_2 & x_3 & \\ \hline 1 & -2 & 3 & b_1 \\ 0 & 1 & -3 & b_2 \\ -2 & 3 & -2 & b_3 \end{array}$$

Use Gauss-Jordan Elimination to reduce it into RREF.

$$\begin{array}{ccc|c} 1 & -2 & 3 & b_1 \\ 0 & 1 & -3 & b_2 \\ -2 & 3 & -2 & b_3 \end{array} \xrightarrow{R_3 \rightarrow R_3 + 2R_1} \begin{array}{ccc|c} 1 & -2 & 3 & b_1 \\ 0 & 1 & -3 & b_2 \\ 0 & -1 & 4 & b_3 + 2b_1 \end{array}$$

Example

$$R_2 \rightarrow R_2 + R_1 \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 3 & b_1 \\ 0 & 1 & -3 & b_2 \\ 0 & 0 & 1 & 2b_1 + b_2 + b_3 \end{array} \right]$$

$$\begin{array}{l} R_1 \rightarrow R_1 - 3R_3 \\ R_2 \rightarrow R_2 + 3R_3 \end{array} \rightarrow \left[\begin{array}{ccc|c} 1 & -2 & 0 & -5b_1 - 3b_2 - 3b_3 \\ 0 & 1 & 0 & 6b_1 + 4b_2 + 3b_3 \\ 0 & 0 & 1 & 2b_1 + b_2 + b_3 \end{array} \right]$$

$$R_1 \rightarrow R_1 + 2R_2 \rightarrow \begin{array}{c} x_1 \quad x_2 \quad x_3 \\ \left[\begin{array}{ccc|c} 1 & 0 & 0 & 7b_1 + 5b_2 + 3b_3 \\ 0 & 1 & 0 & 6b_1 + 4b_2 + 3b_3 \\ 0 & 0 & 1 & 2b_1 + b_2 + b_3 \end{array} \right] \end{array}$$

$$\text{So, } x_1 = 7b_1 + 1b_2 + 3b_3$$

$$x_2 = 6b_1 + 4b_2 + 3b_3$$

$$x_3 = 2b_1 + b_2 + b_3 \quad \star$$

Nature of solutions

Recall:

A system of linear equations that has at least one solution is called **consistent**. A system that has no solution is called **inconsistent**.

Every system of linear equations has either

- (1) no solution,
- (2) exactly one solution or
- (3) infinitely many solutions.

Nature of solutions

After doing the Gaussian Elimination (or Gaussian-Jordan Elimination), look at the augmented matrix in (reduced) row-echelon form:

→ identify all the leading entries

- (1) If the column on the right of the vertical bar is a leading column, then the system of linear equations is inconsistent (have no solution).
- (2) If every leading column is on the left of the vertical bar and each column on the left of the vertical bar is a leading column, then the system of linear equations has a unique solution (have exactly one solution).
- (3) If every leading column is on the left of the vertical bar and there is at least one non-leading column, then the system of linear equations has infinitely many solution. In this case, each of the non-leading column would correspond to one arbitrary parameter.

Nature of solutions

Example:

For each of the following augmented matrices, determine if they have

- (1) no solution,
- (2) exactly one solution, or
- (3) infinitely many solution.

(a) $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 3 & 6 \end{array} \right]$ (2)

(c) $\left[\begin{array}{ccc|c} 0 & 1 & 2 & 3 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 6 \end{array} \right]$ (1)

(e) $\left[\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 5 \\ 0 & 0 & 1 & 2 & 3 \end{array} \right]$ (3)

(b) $\left[\begin{array}{ccc|c} 3 & 2 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 4 & 6 \end{array} \right]$ (2)

(d) $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$ (3)
 x_2 is arbitrary parameter.

(f) $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]$ (2)

Nature of solutions

Example:

Determine whether the following system of linear equations is consistent or inconsistent:

$$3x_1 + 2x_2 + x_3 = 3,$$

$$2x_1 + x_2 + x_3 = 0,$$

$$6x_1 + 2x_2 + 4x_3 = 6.$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ \rightarrow 2 & 1 & 1 & 0 \\ \rightarrow 6 & 2 & 4 & 6 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 - \frac{2}{3}R_1} \left[\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 0 & -\frac{1}{3} & \frac{1}{3} & -2 \\ 0 & -2 & 2 & 0 \end{array} \right]$$

Note that the column
at the right of the vertical
bar is a leading column, the system
is inconsistent

$$\xrightarrow{R_3 \rightarrow R_3 - 6R_2} \left[\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 0 & -\frac{1}{3} & \frac{1}{3} & -2 \\ \rightarrow 0 & 0 & 0 & 12 \end{array} \right]$$

($0x_1 + 0x_2 + 0x_3 = 12$ in third row)

Nature of solutions

Example:

What is the condition that must be satisfied by a , b and c so ^{that} ~~the~~ the following system of linear equations is consistent?

$$x + 2y - 3z = a,$$

$$2x + 6y - 11z = b,$$

$$x - 2y + 7z = c.$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} \textcircled{1} & 2 & -3 & a \\ \rightarrow 2 & 6 & -11 & b \\ \rightarrow 1 & -2 & 7 & c \end{array} \right] \xrightarrow[R_2 \rightarrow R_2 - 2R_1]{R_3 \rightarrow R_3 - R_1} \left[\begin{array}{ccc|c} \textcircled{1} & 2 & -3 & a \\ 0 & \textcircled{2} & -5 & b-2a \\ 0 & -4 & 10 & c-a \end{array} \right]$$

The system is consistent

if and only if

$$c - a + 2(b - 2a) = 0$$

$$\text{or } 2b + c - 5a = 0.$$

$$\xrightarrow{R_3 \rightarrow R_3 + 2R_2} \left[\begin{array}{ccc|c} \textcircled{1} & 2 & -3 & a \\ 0 & \textcircled{2} & -5 & b-2a \\ 0 & 0 & 0 & c-a+2(b-2a) \end{array} \right]$$

We do not want it be a ^{leading} _{entry} ?

Nature of solutions

Example:

Determine the values of k so that the following system of linear equations

$$x + 2y + z = 1,$$

$$2x + ky + 2z = 2,$$

$$4x + 8y + k^2z = 2k.$$

has (a) no solution (b) a unique solution (c) infinitely many solutions.

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & k & 2 & 2 \\ 4 & 8 & k^2 & 2k \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - 4R_1]{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & k-4 & 0 & 0 \\ 0 & 0 & k^2-4 & 2k-4 \end{array} \right]$$

(a) The system has no solution if $k^2 - 4 = 0$ and $2k - 4 \neq 0$, that is $k = -2$.

(b) The system has a unique solution if $k - 4 \neq 0$ and $k^2 - 4 \neq 0$, that is $k \neq \pm 2$ and $k \neq 4$.

(c) The system has infinitely many solutions if $k - 4 = 0$ or both $k^2 - 4 = 0$ and $2k - 4 = 0$, that is $k = 4$ or $k = 2$.

Nature of solutions

Example:

Considering the row-echelon form of the following augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 1 & k & 2 & 5 \\ 2 & k+1 & k^2-1 & k+7 \end{array} \right] \xrightarrow[\text{Elimination}]{\text{Gaussian}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & k-1 & 1 & 1 \\ 0 & 0 & k^2-4 & k-2 \end{array} \right]$$

Determine the values of k so that the following system of linear equations

$$x + y + z = 4,$$

$$x + ky + 2z = 5,$$

$$2x + (k+1)y + (k^2-1)z = k+7.$$

has (a) no solution (b) a unique solution (c) infinitely many solutions.

the cases that we need to consider are $k-1=0$ and $k^2-4=0$
 $k=1$ $k=\pm 2$

When $k \neq 1$, the augmented matrix in row-echelon form becomes

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -3 & -1 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + 3R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 2 \end{array} \right] \leftarrow \text{no solution.}$$

Nature of solutions

When $k = -2$, the augmented matrix becomes

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & -3 & 1 & 1 \\ 0 & 0 & 0 & -4 \end{array} \right] \leftarrow \text{no solution}$$

When $k = 2$, the augmented matrix becomes

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 4 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \leftarrow \text{infinitely many solutions}$$

- a) The system has no solution when $k=1$ or $k=-2$
- b) The system has unique solution when $k \neq 1$ or $k \neq -2$
- c) The system has infinitely many solutions when $k=2$.

Nature of solutions

Example:

Using the following steps of reducing the matrix to row-echelon form, or otherwise, answer the questions below.

$$\begin{aligned} & \left[\begin{array}{ccc|c} k & 1 & 1 & -k^3 - 5k^2 - 5k + 10 \\ 1 & k & 1 & -k^2 \\ 2 & 1 & 4k & -k \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_1} \\ & \left[\begin{array}{ccc|c} 1 & k & 1 & -k^2 \\ k & 1 & 1 & -k^3 - 5k^2 - 5k + 10 \\ 2 & 1 & 4k & -k \end{array} \right] \xrightarrow{\begin{array}{l} R_2 \rightarrow R_2 - kR_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array}} \\ & \xrightarrow{\text{blue arrow}} \left[\begin{array}{ccc|c} 1 & k & 1 & -k^2 \\ 0 & 1 - k^2 & 1 - k & -5k^2 - 5k + 10 \\ 0 & 1 - 2k & 4k - 2 & -k + 2k^2 \end{array} \right] \xrightarrow{\begin{array}{l} R_2 \rightarrow \frac{1}{1-k} R_2 \\ R_3 \rightarrow \frac{1}{2k-1} R_3 \end{array}} \quad \begin{array}{l} \text{red arrow } k \neq 1 \\ \text{red arrow } k \neq \frac{1}{2} \end{array} \\ & \left[\begin{array}{ccc|c} 1 & k & 1 & -k^2 \\ 0 & 1 + k & 1 & 5k + 10 \\ 0 & -1 & 2 & k \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & k & 1 & -k^2 \\ 0 & -1 & 2 & k \\ 0 & 1 + k & 1 & 5k + 10 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + (1+k)R_2} \\ & \left[\begin{array}{ccc|c} \textcircled{1} & k & 1 & -k^2 \\ 0 & \textcircled{-1} & 2 & k \\ 0 & 0 & \underline{2k+3} & \underline{k^2+6k+10} \end{array} \right] \end{aligned}$$

Nature of solutions

Determine the values of k so that the following system of linear equations

$$kx + y + z = -k^3 - 5k^2 - 5k + 10,$$

$$x + ky + z = -k^2,$$

$$2x + y + 4kz = -k.$$

has (a) no solution (b) infinitely many solutions (c) a unique solution.

or The system has no solution when $2k-3=0$ and $k^2+6k+10 \neq 0$,

that is $k = \frac{-3}{2}$

$$\text{Let } A = \left[\begin{array}{ccc|c} 1 & k & 1 & -k^2 \\ 0 & 1-k & 1-k & -5k^2-5k+10 \\ 0 & 1-2k & 4k-2 & -k+2k^2 \end{array} \right]$$

When $k=1$,

$$A = \left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & 3 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & -1 \\ 0 & -1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

infinitely many solutions

When $k = \frac{1}{2}$

$$A = \left[\begin{array}{ccc|c} 1 & \frac{1}{2} & 1 & -\frac{1}{4} \\ 0 & \frac{3}{4} & \frac{1}{2} & \frac{25}{4} \\ 0 & 0 & 0 & 0 \end{array} \right] \leftarrow \text{infinitely many solutions}$$

The system has infinitely many solutions when $k=1$ or $k=\frac{1}{2}$.

c) The system has unique solution when $k \neq \frac{-3}{1}, 1, \frac{1}{2}$.

Homogeneous system ($B = 0$)

A system of m linear equations in n unknowns $x_1, x_2, \dots, x_n$ is said to be **homogeneous** if it has the form

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0$$

$$\vdots$$

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0$$

where a_{ij} , $1 \leq i \leq m$, $1 \leq j \leq n$ are real constants.

An homogeneous system always has a solution, $x_1 = 0, x_2 = 0, x_3 = 0, \dots, x_n = 0$. This solution is called the **trivial solution**. Any other solution, if it exists, is called **non-trivial solution**.

A homogeneous system of linear equations is always consistent. It has either only the trivial solution or infinitely many (non-trivial) solutions in addition to the trivial solution.

A homogeneous system of linear equations with more unknowns than equations has infinitely many solutions.

Homogeneous system ($B = 0$)

Example:

Determine the value(s) of k such that

$$x - 5y + 3z = 0,$$

$$5x - y - kz = 0,$$

$$x - 2y + kz = 0.$$

has nontrivial solution. *← has infinitely many solutions*

The augmented matrix for the system is

$$\left[\begin{array}{ccc|c} 1 & -5 & 3 & 0 \\ 5 & -1 & -k & 0 \\ 1 & -2 & k & 0 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - 5R_1} \left[\begin{array}{ccc|c} 1 & -5 & 3 & 0 \\ 0 & 24 & -k-15 & 0 \\ 0 & 3 & k-3 & 0 \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow 8R_3} \left[\begin{array}{ccc|c} 1 & -5 & 3 & 0 \\ 0 & 24 & -k-15 & 0 \\ 0 & 24 & 8k-24 & 0 \end{array} \right]$$

$$\xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & -5 & 3 & 0 \\ 0 & 24 & -k-15 & 0 \\ 0 & 0 & 9k-9 & 0 \end{array} \right]$$

The system has non-trivial solutions if $9k-9=0$, that is $k=1$.

If $k \neq 1$, then the system has only trivial solution

$$x=y=z=0.$$

↖ a unique solution.